

David You

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EDUCATION

Northeastern University

Boston, MA

Candidate for B.S. in Mechanical Engineering GPA: 3.636/4.0, Dean's List

Expected Graduation: May 2029

- Clubs & Activities: Northeastern Electric Racing, American Society of Mechanical Engineers, Lead360, Little Brothers – Friends of the Elderly

EXPERIENCE

Research Assistant

Boston, MA

W. M. Keck Laboratory for Integrated Ferroics

May 2026 – Present

- Engineered inductance coil assemblies and validated field output with a multimeter using statistical error analysis
- Characterized nickel-zinc ferrite cores across a 10 kHz–1 MHz sweep to identify the lowest-loss configuration
- Quantified the inductance-versus-turns relationship across 10+ configurations, confirming the expected N^2 dependence and extracting core permeability from the linearized fit

EXP Explorer

Boston, MA

Makerspace at EXP

July 2026 – August 2026

- Operated SLA printing, woodworking handheld power tools, and industrial sewing machines, expanding fabrication support across multiple workflows
- Managed equipment readiness and shop workflow, troubleshooting faults to minimize downtime

Philips

Andover, MA

Engineering Intern

April 2025 – August 2025

- Contributed to R&D for next-generation patient monitoring systems, analyzing 15+ feature enhancements and system requirements to drive measurable improvements
- Validated 50+ design requirements and reviewed compliance protocols across 3 FDA submission packages in a medical device R&D workflow
- Collaborated with a cross-functional team of 8 engineers to streamline documentation processes, reducing review cycle time by 20%

Town Pantry

Winchester, MA

Customer Service Associate

March 2024 – Present

- Delivered service in a high-volume store, processing 100+ transactions per shift with 99% accuracy while handling \$2,000+ daily with zero discrepancies
- Trained 3 new team members on POS systems, inventory protocols, and customer service practices

PROJECTS

Borealga

January 2026 – April 2026

- Engineered a spring-powered pickleball cannon to 100% launch reliability across 30+ shots, designing a PETG plunger and a 30.42 lb compression spring
- Built an IR break-beam target on a Raspberry Pi Pico (MicroPython) with sub-500 ms LED feedback, engaging 70+ children at Boston Children's Museum's Fun Labs

OverIDE

September 2025 – December 2025

- Led a semester-long capstone team, producing CAD models, drawings, and documentation for 25+ components in AutoCAD and reporting status, schedule, and risk to stakeholders
- Integrated 5 hardware subsystems with 500+ lines of Python under Git-based configuration control, developing verification protocols to achieve 95% system response reliability

Arduino Autonomous Navigation Robot

July 2024 – August 2024

- Designed and 3D-printed 8 custom airframe and mounting components through 15+ iterations, evaluating materials and tolerances with DFM to improve structural stability by 40%
- Built a self-navigating vehicle with an Arduino Uno and ultrasonic sensors, programming 300+ lines of C++ to reach 92% obstacle detection accuracy through structured testing

SKILLS

CAD & Detail Design: SOLIDWORKS (Detail, Assembly, Installation Drawings), AutoCAD, 2D Drawings, GD&T fundamentals, Design for Manufacturing, Tolerance Analysis

Engineering Process: Requirements Trace, Verification & Validation, Configuration Control, Engineering Change Requests, Status & Risk Reporting

Fabrication & Testing: 3D Printing (FDM/SLA), Rapid Prototyping, Material Selection, Structured Testing, Statistical Error Analysis, Soldering, Hand & Power Tools

Software & Tools: MATLAB, Python, C++, MicroPython, Arduino, Raspberry Pi, Microsoft Excel, Git/GitHub **Languages:** English (Fluent), Chinese (Fluent)